Class Eight Science Notes: The Water Cycle

Water moves through the environment in a continuous loop called the water cycle. Energy from the sun drives evapotranspiration, which combines evaporation from oceans, rivers and lakes with transpiration from the leaves of plants. Rising vapour cools and condenses on tiny particles called condensation nuclei, forming cloud droplets. When droplets grow heavy enough they fall as precipitation: rain, hail, sleet or snow. Some water sinks into the soil and becomes groundwater; the rest runs off into streams and returns to the sea.

Key terms: evapotranspiration, condensation nuclei, precipitation, groundwater.

Class Eight Science Notes: Photosynthesis

Green leaves make food using light, water and carbon dioxide. The green pigment chlorophyll inside the chloroplast absorbs mainly red and blue light and reflects green light, which is why leaves look green. The overall equation is six molecules of carbon dioxide plus six molecules of water, in the presence of light and chlorophyll, giving one molecule of glucose and six molecules of oxygen.

Gas exchange happens through tiny pores called stomata on the underside of the leaf. Guard cells open and close the stomata to balance water loss against the carbon dioxide the plant needs.

Key terms: chlorophyll, chloroplast, stomata, guard cells, glucose, oxygen.

Class Nine Mathematics Notes: Quadratic Equations

A quadratic equation has the form ax squared plus bx plus c equals zero, where a is not zero. The discriminant, b squared minus four ac, decides the nature of the roots: positive means two distinct real roots, zero means one repeated real root, and negative means no real roots. The roots can be found by factoring, by completing the square, or with the quadratic formula: minus b plus or minus the square root of the discriminant, all divided by two a.

The graph of a quadratic function is a parabola. When a is positive the parabola opens upward and its vertex is the minimum point; when a is negative it opens downward and the vertex is the maximum. The axis of symmetry is x equals minus b over two a.

Key terms: discriminant, quadratic formula, parabola, vertex, axis of symmetry.